Domingo Ranieri

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Profile

I hold a Master's degree in Theoretical Physics, where I built a strong foundation in Python programming, numerical methods, and data analysis using tools such as Pandas, NumPy, Matplotlib, TensorFlow, Keras, and PyTorch. My studies also strengthened my problem-solving abilities and my capacity to model complex systems, skills that I now apply across advanced data engineering, AI, and distributed computing projects. My experience spans machine learning engineering, blockchain-enhanced data management, federated cloud orchestration, quantum computing, and modern data lakehouse architectures.

Education

Master's Degree in Physics — *University of Bologna*
Theoretical Physics — *2020 - 2022*
Final grade: 110/110 cum laude
Thesis: Simulation of neuromuscular control using a quantum computer

Bachelor's Degree in Physics — *University of Bologna*
Physics — *2017 - 2020*
Final grade: 110/110 cum laude
Thesis: Fractal Universe model and Cosmic Acceleration

Work experience

ICSC — *Mar 2025 - Present*
Software Engineer — *Bologna, Italy*

- Research and development in AI, data engineering, and quantum computing
- Design and implementation of advanced software services
- Delivery of technical courses on AI, cloud, and data engineering
- Participation in conferences and collaborative research projects

ICSC-INFN — *Sep 2023 - Feb 2025*
Research Fellow — *Bologna, Italy*

- Development of AI and blockchain-based services
- Applied research in distributed and intelligent systems
- Participation in workshops and scientific events

Key Projects

INFN Data Cloud — *2024 - 2025*
Federated Cloud Orchestration & AI-Ranker

- Designed and contributed to the architecture of a federated cloud orchestration platform (Indigo-Orchestrator) enabling automated deployment across distributed cloud providers
- Developed the AI-Ranker component to optimize deployment decisions based on dynamic metrics
- Built and managed machine learning pipelines using **scikit-learn** and **MLflow** (experiment tracking, model versioning, deployment)

- Implemented real-time data ingestion pipelines with **Apache Kafka** and feature engineering workflows with **Pandas** and **NumPy**
- Delivered end-to-end integration from data ingestion to model inference, improving system scalability, automation, and decision accuracy

Domain-Specific RAG System *2026 - Present*
Genetic Research

- Designed a **domain-adapted Retrieval-Augmented Generation (RAG)** system for querying and synthesizing biomedical literature
- Implemented hybrid retrieval strategies combining **vector search** and **GraphRAG**
- Built scalable multi-source data pipelines using **Milvus** for vector storage
- Developed an **agentic RAG architecture** capable of adapting retrieval strategies based on query intent (exploratory vs evidence-based)
- Improved traceability and reliability of generated outputs through evidence-grounded responses and source attribution

Multi-Agent Chatbot *2026 - Present*
Genetic Pathogenicity Analysis

- Architected a **multi-agent AI system** for genetic variant interpretation and pathogenicity analysis
- Designed modular agent workflows with task delegation and reasoning pipelines using **CrewAi**
- Integrated **Model Context Protocol (MCP)** for standardized tool interaction and context sharing across agents
- Developed custom tools for real-time data analysis, including API integration, automated workflows, and scientific data processing
- Enabled generation of analytical outputs (charts, reports) for decision support and research workflows

Technical skills

Programming	Python, C++, Go, Bash
ML/AI	TensorFlow, PyTorch, Keras, Scikit-learn
Data Engineering	Kafka, Airflow, Iceberg, Trino, MinIO
Cloud & DevOps	Docker, Kubernetes, Ansible
Databases	PostgreSQL, MongoDB, MySQL
Web	Flask, FastAPI, HTML, CSS, JavaScript
Tools	MLflow, OpenMetadata, Great Expectations, Git

Languages

Italian	Native
English	B2

Communication and interpersonal skills

Effective presentation and public speaking

Strong technical writing and documentation skills

Collaborative teamwork in multidisciplinary environments

Ability to explain complex concepts clearly